

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression, SGDRegressor
from sklearn.preprocessing import PolynomialFeatures, StandardScaler
from sklearn.metrics import r2_score
from sklearn.pipeline import Pipeline
```

```
In [16]: df = pd.read_csv("Electric_consumption_dataset/train_energy_data.csv")
```

```
In [17]: X.head()
```

```
Out[17]:
```

	Building Type	Square Footage	Number of Occupants	Appliances Used	Average Temperature	Day of Week	Cost
0	Residential	7063	76	10	29.84	Weekday	
1	Commercial	44372	66	45	16.72	Weekday	
2	Industrial	19255	37	17	14.30	Weekend	
3	Residential	13265	14	41	32.82	Weekday	
4	Commercial	13375	26	18	11.92	Weekday	

```
In [18]: y = pd.read_csv("Electric_consumption_dataset/test_energy_data.csv")
```

```
In [19]: y.head()
```

```
Out[19]:
```

	Building Type	Square Footage	Number of Occupants	Appliances Used	Average Temperature	Day of Week	Cost
0	Residential	24563	15	4	28.52	Weekday	
1	Commercial	27583	56	23	23.07	Weekend	
2	Commercial	45313	4	44	33.56	Weekday	
3	Residential	41625	84	17	27.39	Weekend	
4	Residential	36720	58	47	17.08	Weekday	

```
In [38]: lr = LinearRegression()
```

```
In [39]: X_train = df.iloc[:, :-1]
```

```
In [40]: X_train.head()
```

Out[40]:

	Building Type	Square Footage	Number of Occupants	Appliances Used	Average Temperature	Day of Week
0	Residential	7063	76	10	29.84	Weekday
1	Commercial	44372	66	45	16.72	Weekday
2	Industrial	19255	37	17	14.30	Weekend
3	Residential	13265	14	41	32.82	Weekday
4	Commercial	13375	26	18	11.92	Weekday

In [51]: `X_train.drop(columns = ["Building Type"], axis=1, inplace=True)`

In [52]: `X_train.head()`

Out[52]:

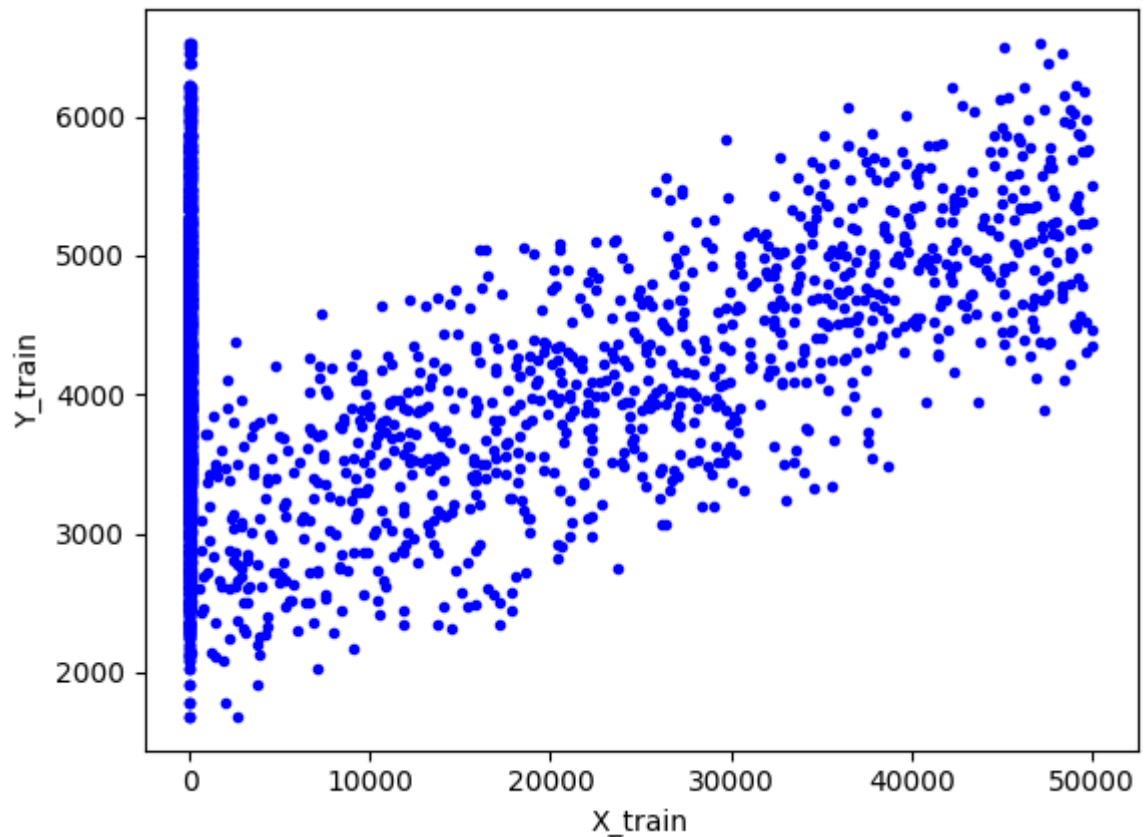
	Square Footage	Number of Occupants	Appliances Used	Average Temperature
0	7063	76	10	29.84
1	44372	66	45	16.72
2	19255	37	17	14.30
3	13265	14	41	32.82
4	13375	26	18	11.92

In [53]: `Y_train = df["Energy Consumption"]`

In [54]: `Y_train.info()`

```
<class 'pandas.core.series.Series'>
RangeIndex: 1000 entries, 0 to 999
Series name: Energy Consumption
Non-Null Count  Dtype
-----  -----
1000 non-null   float64
dtypes: float64(1)
memory usage: 7.9 KB
```

In [59]: `plt.plot(X_train, Y_train, 'b.')`
`plt.xlabel("X_train")`
`plt.ylabel("Y_train")`
`plt.show()`



```
In [55]: lr.fit(X_train,Y_train)
```

```
Out[55]: ▼ LinearRegression ⓘ ?
          ► Parameters
```

```
In [68]: df2 = pd.read_csv("Electric_consumption_dataset/test_energy_data.csv")
```

```
In [69]: X_test = df2.iloc[:, :-1]
```

```
In [70]: X_test.drop(columns = ["Building Type"], axis=1, inplace=True)
```

```
In [72]: X_test.head()
```

```
Out[72]:
```

	Square Footage	Number of Occupants	Appliances Used	Average Temperature	Day of Week
0	24563	15	4	28.52	Weekday
1	27583	56	23	23.07	Weekend
2	45313	4	44	33.56	Weekday
3	41625	84	17	27.39	Weekend
4	36720	58	47	17.08	Weekday

```
In [73]: X_test.drop(columns = ["Day of Week"], axis=1, inplace=True)
```

```
In [74]: X_test.head()
```

```
Out[74]:
```

	Square Footage	Number of Occupants	Appliances Used	Average Temperature
0	24563	15	4	28.52
1	27583	56	23	23.07
2	45313	4	44	33.56
3	41625	84	17	27.39
4	36720	58	47	17.08

```
In [76]: Y_test = df2["Energy Consumption"]
```

```
In [77]: Y_test.info()
```

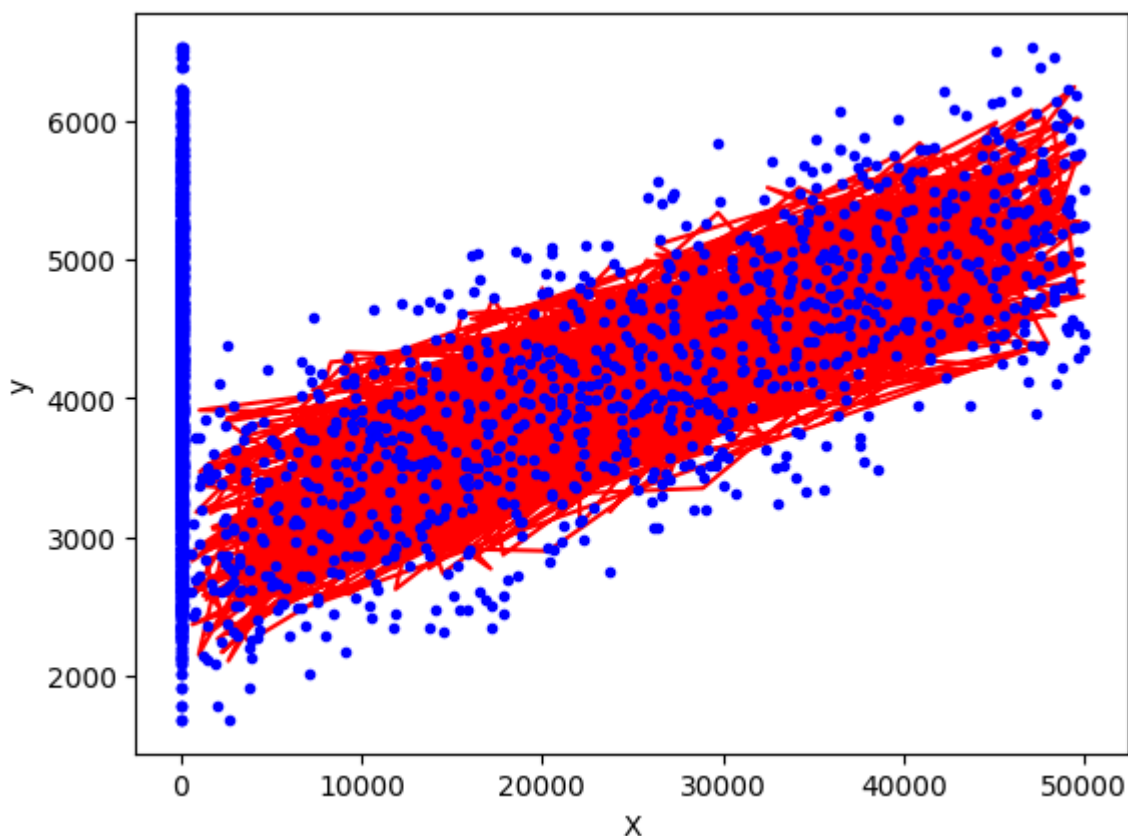
```
<class 'pandas.core.series.Series'>
RangeIndex: 100 entries, 0 to 99
Series name: Energy Consumption
Non-Null Count  Dtype
-----  -----
100 non-null    float64
dtypes: float64(1)
memory usage: 932.0 bytes
```

```
In [78]: Y_pred = lr.predict(X_test)
```

```
In [81]: r2_score(Y_test, Y_pred)
```

```
Out[81]: 0.7307243739551794
```

```
In [89]: plt.plot(X_train, lr.predict(X_train), color = 'r')
plt.plot(X_train, Y_train, "b.")
plt.xlabel("X")
plt.ylabel("y")
plt.show()
```



```
In [90]: poly = PolynomialFeatures(degree=2, include_bias=True)
X_train_trans = poly.fit_transform(X_train)
X_test_trans = poly.transform(X_test)
```

```
In [91]: poly.powers_
```

```
Out[91]: array([[0, 0, 0, 0],
 [1, 0, 0, 0],
 [0, 1, 0, 0],
 [0, 0, 1, 0],
 [0, 0, 0, 1],
 [2, 0, 0, 0],
 [1, 1, 0, 0],
 [1, 0, 1, 0],
 [1, 0, 0, 1],
 [0, 2, 0, 0],
 [0, 1, 1, 0],
 [0, 1, 0, 1],
 [0, 0, 2, 0],
 [0, 0, 1, 1],
 [0, 0, 0, 2]])
```

```
In [94]: print(X_train)
print(X_train_trans)
```

	Square Footage	Number of Occupants	Appliances Used	Average Temperature
0	7063	76	10	29.84
1	44372	66	45	16.72
2	19255	37	17	14.30
3	13265	14	41	32.82
4	13375	26	18	11.92
...	...	...	...	...
995	14419	68	44	23.95
996	12194	7	22	14.67
997	39562	88	20	32.18
998	8348	67	37	16.48
999	15813	57	11	31.40

```
[1000 rows x 4 columns]
[[1.0000000e+00 7.0630000e+03 7.6000000e+01 ... 1.0000000e+02
 2.9840000e+02 8.9042560e+02]
[1.0000000e+00 4.4372000e+04 6.6000000e+01 ... 2.0250000e+03
 7.5240000e+02 2.7955840e+02]
[1.0000000e+00 1.9255000e+04 3.7000000e+01 ... 2.8900000e+02
 2.4310000e+02 2.0449000e+02]
...
[1.0000000e+00 3.9562000e+04 8.8000000e+01 ... 4.0000000e+02
 6.4360000e+02 1.0355524e+03]
[1.0000000e+00 8.3480000e+03 6.7000000e+01 ... 1.3690000e+03
 6.0976000e+02 2.7159040e+02]
[1.0000000e+00 1.5813000e+04 5.7000000e+01 ... 1.2100000e+02
 3.4540000e+02 9.8596000e+02]]
```

```
In [96]: lr = LinearRegression()
lr.fit(X_train_trans,Y_train)
```

```
Out[96]: ▼ LinearRegression ⓘ ?
```

```
► Parameters
```

```
In [97]: y_pred = lr.predict(X_test_trans)
```

```
In [100... r2_score(Y_test,y_pred)
```

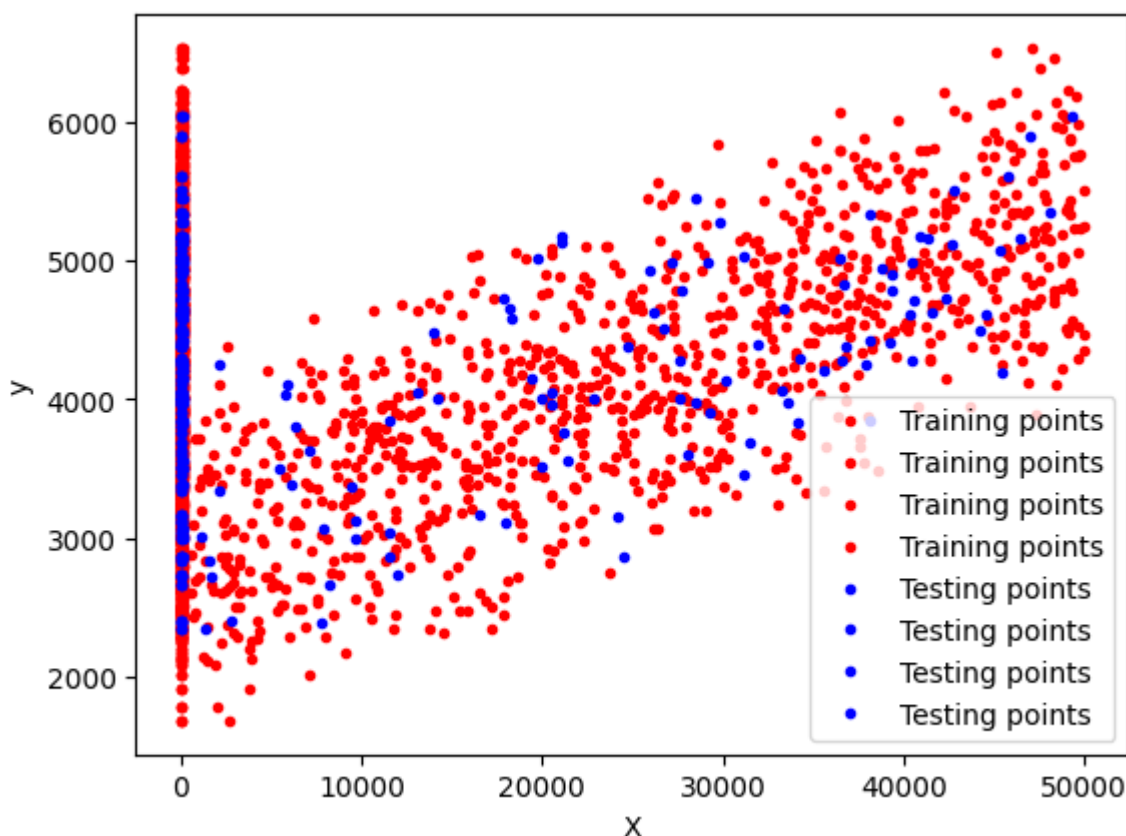
```
Out[100... 0.7271503974241591
```

```
In [101... print(lr.coef_)
```

```
print(lr.intercept_)
```

```
[ 0.00000000e+00  4.59150779e-02  8.93644510e+00  3.20930964e+01
 -2.48403340e+01 -1.47051263e-08  2.72430251e-05 -1.10746861e-05
  1.72876594e-04  7.86896856e-03 -4.00432146e-02  4.18190948e-02
 -1.50218341e-01 -6.88138826e-02  3.25264752e-01]
2154.5159120127673
```

```
In [105... plt.plot(X_train, Y_train, "r.", label='Training points')
plt.plot(X_test, Y_test, "b.", label='Testing points')
plt.xlabel("X")
plt.ylabel("y")
plt.legend()
plt.show()
```



```
In [ ]:
```